

Problem Statement

Smallholder farmers and agricultural extension agents in local communities face a significant challenge in accessing timely, localized, and actionable agronomic advice. While a wealth of information exists in research papers, government bulletins, and agricultural guides, this knowledge is often buried in dense, technical language and dispersed across many different sources. Farmers cannot easily ask a question about a specific pest or weather pattern and get a clear, concise answer. The "last mile" of information delivery is broken, leaving farmers to make critical decisions without the best available science, which can lead to crop failure and lost income.

Who is affected by this problem?

(Who experiences the problem?)

The primary people affected are smallholder farmers, particularly those in rural and peri-urban areas, who lack direct access to agricultural experts. Their families are also affected, as their food security and income are directly tied to the farm's success. Furthermore, local agricultural extension agents are affected; they are often under-resourced and struggle to provide personalized support to every farmer in their region, leading to stress and burnout.

Why is solving this problem important?

If this problem remains unsolved, farmers will continue to make decisions based on outdated practices or incomplete information. This leads to lower crop yields, increased vulnerability to pests and diseases, and inefficient use of resources like water and fertilizers. The economic consequences are severe, trapping farming families in a cycle of low productivity and poverty. At a larger scale, food insecurity persists, and communities remain less resilient to climate change. It also means the vast knowledge generated by agronomic research remains largely unused, representing a significant loss of potential human benefit.

The core values guiding our solution will be:

1. Knowledge Equity: Democratizing access to information so that all farmers, regardless of their education or resources, can benefit from expert knowledge.
2. Economic Livelihood: Supporting the financial well-being and resilience of farming families.
3. Food Security: Contributing to a stable and sufficient food supply for the community.
4. Cultural Preservation: Respecting and integrating local and indigenous farming knowledge.
5. Environmental Sustainability: Promoting practices that protect the land and resources for future generations.

We chose these values because they are at the heart of the issue. A solution that doesn't prioritize knowledge equity could exacerbate the "digital divide," helping only those who are already more privileged. The entire purpose of the solution is to improve economic livelihood and food security; these are the human outcomes we are working towards. We must be careful not to ignore local wisdom; the goal is to augment, not replace, a farmer's knowledge, thus upholding cultural

preservation. Finally, any advice given must be sustainable to ensure long-term benefits, which is why environmental sustainability is a critical guiding principle.

The primary ethical risks include:

1. **Information Bias:** The system might only retrieve advice from specific, dominant sources, ignoring locally relevant or indigenous knowledge, which could lead to impractical recommendations.
2. **Digital Divide:** A solution that requires high-end smartphones or expensive data plans could be inaccessible to the very farmers it aims to help.
3. **Over-Reliance:** Farmers might blindly trust the AI's advice, potentially ignoring their own practical experience or the specific conditions of their field.
4. **Data Privacy:** If the system logs user queries, the data might contain sensitive information about a farmer's specific practices or challenges.

To mitigate these risks, we will:

1. **Curate a Diverse Knowledge Base:** We will actively seek to include community-generated and indigenous farming knowledge alongside formal research.
2. **Design for Accessibility:** The interface will be optimized for mobile devices and low-bandwidth environments, using natural language voice or text inputs.
3. **Communicate Uncertainty:** The system will clearly present the source of its information and its confidence level, explicitly encouraging users to use their own judgment.
4. **Prioritize Data Privacy:** User queries and data will be anonymized and stored securely. We will be transparent about what data is collected and why.

Success is defined by the real-world impact on our community. Key outcomes include:

1. **Improved Farmer Decision-Making:** Farmers report that the advice they receive helps them solve problems and improve their practices.
2. **Increased Productivity:** Measurable improvements in crop yields or reduction in crop losses.
3. **Enhanced Agent Efficiency:** Extension agents can serve more farmers more effectively.
4. **High User Trust:** Farmers and agents report that they find the system helpful, reliable, and relevant to their context.
5. **Adoption of Sustainable Practices:** An increase in the use of environmentally sound techniques.

Problem Statement

Smallholder farmers in our community struggle to access timely, localized agronomic advice due to the technical and dispersed nature of available information. This creates a significant gap between agricultural research and on-the-ground practice, leading to lower yields, economic instability, and food insecurity. Our project aims to bridge this gap by developing an accessible advisory system that can retrieve and present the most relevant information from a vast repository of documents. Our solution will be guided by values of knowledge equity, economic livelihood, and environmental sustainability, ensuring that all farmers, regardless of their background, can benefit. By empowering farmers with better information, we aim to enhance decision-making, improve food security, and build a more resilient and prosperous community.